

Wenhao Peng

RF Acoustic Wave Devices

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Biography

Wenhao Peng is a Staff Electrical Engineer at Skyworks Solutions, specializing in RF acoustic filter technology. He develops device models and simulation methods that support filter design across BAW and SAW technologies. His work connects thermodynamics and mechanical vibrations with practical resonator and filter design. In ferroelectric BAW technology, he developed a physics-based equivalent-circuit framework that bridges nonlinear ferroelectric thermodynamics and measured resonator behavior. In piezoelectric FBAR technology, he established an analytical relation between field-based and resonance-frequency-based definitions of electromechanical coupling, making modal field distributions directly useful for evaluating layer stacks and device performance. At Skyworks, he is developing modal and transfer-matrix approaches to accelerate established finite-element models for both BAW and SAW and integrate them into the filter-design workflow. He also demonstrated the first periodically poled AlN/ScAlN/AlN trilayer FBAR operating above 50 GHz. He received the Ph.D. degree in electrical and computer engineering from the University of Michigan, Ann Arbor, in 2026.

Selected Contributions

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| Modal analysis | Proved the analytical relation between the two standard definitions of electromechanical coupling (k_t^2): the Berlincourt form computed from acoustic field distributions, and the form determined from the series and parallel resonance frequencies. Derived the relation through modal analysis of the open-circuit eigenmodes of an FBAR. The result makes acoustic field plots quantitatively interpretable, showing what a given mode contributes to resonator coupling, and enables direct evaluation of modal contributions to layer stack performance, reducing reliance on iterative optimization against simulated admittance. This relation was applied in both mm-wave FBAR papers, with the derivation presented at IMS 2024 and IMS 2025, and now informs work on accelerating finite element simulation for the filter design workflow. |
| Finite element methods | Adapted interpolation methods from finite element analysis into the thermodynamic modeling of ferroelectrics, bridging a material level thermodynamic description and a compact circuit representation of acoustic resonance. Produced the first equivalent-circuit model of ferroelectric-driven BAW resonators derived from first principles rather than empirical fitting, reproducing measured response across DC bias. First author, IEEE TMTT 2022 and IMS 2021. |
| mm-Wave scaling | First periodically poled AlN/ScAlN/AlN trilayer FBAR operating above 50 GHz, at 56 GHz. The modal-analysis result guided electrode and layer-stack selection for a second-generation 36 GHz device. First author of the corresponding IMS 2024 and IMS 2025 papers. Built on co-demonstrated controlled ferroelectric polarization switching in ultrawide-bandgap AlN/ScAlN multilayers (Applied Physics Letters 2023). |
| Switchless frontends | Co-developed the first switchless quad-band filter bank based on ferroelectric BST FBARs. The work also resulted in the first intrinsically switchable balanced ferroelectric FBAR filter and the first RF acoustic phase-inversion switch based on poled ferroelectrics. |

Experience

- Feb 2026–Present **RF Acoustic Staff Electrical Engineer**, *Skyworks Solutions, Inc.*, Irvine, CA.
- Provide device-modeling and simulation support for BAW and SAW filter design in higher-frequency cellular bands.
 - Develop modal and transfer-matrix methods to accelerate established finite-element acoustic models for both BAW and SAW and integrate them into the filter-design workflow.
 - Diagnose resonator and filter performance issues from design through production using equivalent-circuit and multiphysics models.
- Sep 2019–Dec 2025 **Graduate Student Research Assistant**, *University of Michigan*, Ann Arbor, MI.
Advisor: Prof. Amir Mortazawi
- 2025 *mm-Wave acoustic resonators based on periodically poled III-nitride thin films*
- Extended periodically poled AlN/ScAlN/AlN trilayer FBARs above 50 GHz, demonstrating the first such device at 56 GHz, followed by a second-generation 36 GHz resonator with a redesigned electrode and layer stack.
 - Established the theoretical basis for the mm-wave device work by proving the relation between the Berlincourt and resonance-frequency definitions of electromechanical coupling. By evaluating layer contributions on the modal coordinates, the layer stack design process becomes quantitative about how material elastic, piezoelectric, and dielectric properties interact at the target resonance.
 - Advanced the nanofabrication and pulsed-field polarization-switching processes of MBE grown AlN/ScAlN multilayer structures.
 - Performed mm-wave characterization to 67 GHz with on-wafer de-embedding structures validated in ADS Momentum and Ansys HFSS.
- Reconfigurable RF frontends based on intrinsically switchable BST acoustic resonators*
- Developed the first equivalent-circuit model of ferroelectric-driven BAW resonators based on ferroelectric thermodynamics. Adapted finite element interpolation methods to bridge the material thermodynamic description and a compact circuit representation of acoustic resonance. Validated the model by direct comparison with measurements across DC bias.
 - Contributed to the first switchless quad-band BST FBAR filter bank and the first intrinsically switchable balanced ferroelectric FBAR filter at 2 GHz.
 - Contributed to the first RF acoustic phase inversion switch based on poled-ferroelectric stacked crystal filters.
 - Synthesized filters using the image parameter method, applying Mason and mBVD resonator models, and carried the designs through complete cleanroom fabrication and DC-bias-dependent characterization.

Education

- 2026 **Ph.D., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Dissertation: *Layered Ferroelectric Millimeter Wave BAW Resonators: Modeling and Validation based on On-Wafer Characterization*. Advisor: Prof. Amir Mortazawi.
- 2024 **M.S.E., Mechanical Engineering**, *University of Michigan*, Ann Arbor, MI.
Dynamics and vibrations
- 2019 **M.S., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Integrated circuits and VLSI
- 2018 **B.S.E., Electrical Engineering**, *University of Michigan*, Ann Arbor, MI.
Summa Cum Laude

2018 **B.S.E., Electrical and Computer Engineering**, *Shanghai Jiao Tong University*, Shanghai, China.
Dual degree with U-M

Technical Skills

Devices & materials	BAW/FBAR resonator and filter design; thin-film piezoelectrics and ferroelectrics (AlN, ScAlN, BST); periodically poled III-nitride multilayers; MEMS; RF frontend architectures
Modeling	Mason and mBVD equivalent circuits; thermodynamic modeling of ferroelectric nonlinearity; modal analysis; image-parameter filter synthesis; multiphysics FEM (COMSOL); EM co-simulation (ADS Momentum, Ansys HFSS); structural vibration analysis
Fabrication	Cleanroom process integration; photolithography; physical vapor deposition; wet/dry/plasma etching; thermal oxidation; device processing compatible with MBE-grown films; pulsed-field poling
Measurement	On-wafer S-parameter characterization to 67 GHz; GSG probing; VNA calibration and de-embedding; DC-bias-dependent RF characterization; material parameter extraction
Tools	ADS, COMSOL Multiphysics, Ansys HFSS, MATLAB, Python, Cadence
AI-assisted work	LLM-assisted Python development, code review, numerical workflow automation, and technical documentation

Journal Publications

- 2025 H. Desai, **W. Peng**, and A. Mortazawi, "Single-Pole Single-Throw RF Acoustic Phase Inversion Switch Leveraging Poled Ferroelectrics," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 1, pp. 6–13, 2025.
- 2023 D. Wang, P. Wang, S. Mondal, J. Liu, M. Hu, M. He, S. Nam, **W. Peng**, S. Yang, D. Wang, Y. Xiao, Y. Wu, A. Mortazawi, and Z. Mi, "Controlled Ferroelectric Switching in Ultrawide Bandgap AlN/ScAlN Multilayers," *Appl. Phys. Lett.*, vol. 123, no. 10, p. 103506, 2023.
- 2023 S. Nam, **W. Peng**, P. Wang, D. Wang, Z. Mi, and A. Mortazawi, "A mm-Wave Trilayer AlN/ScAlN/AlN Higher Order Mode FBAR," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 6, pp. 803–806, 2023.
- 2022 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Phenomenological Circuit Modeling of Ferroelectric-Driven Bulk Acoustic Wave Resonators," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 1, pp. 919–925, 2022.
- 2021 S. Nam, M. Z. Koohi, **W. Peng**, and A. Mortazawi, "A Switchless Quad Band Filter Bank Based on Ferroelectric BST FBARs," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 6, pp. 662–665, 2021.

Conference Publications

- 2025 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 36 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, San Francisco, CA, 2025, pp. 786–789. *(presented by author)*
- 2024 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 56 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, Washington, DC, 2024, pp. 150–153. *(presented by author)*
- 2021 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, “Physics Based Modeling of Electrostriction Based BAW Resonators,” *IEEE MTT-S International Microwave Symposium (IMS)*, Atlanta, GA, 2021, pp. 214–217. *(presented by author)*
- 2020 M. Z. Koohi, **W. Peng**, and A. Mortazawi, “An Intrinsically Switchable Balanced Ferroelectric FBAR Filter at 2 GHz,” *IEEE/MTT-S International Microwave Symposium (IMS)*, 2020, pp. 131–134.
- 2018 Y. Dai, **W. Peng**, Y. Wang, L.-X. Chuo, K. Suri, H. Zheng, D. Wentzloff, and H.-S. Kim, “Implementation and Evaluation of Bi-Directional WiFi Back-Channel Communication,” *IEEE 29th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2018, pp. 1–7.

Peer Review Contributions

- 49 reviews **IEEE:** J. Microelectromechanical Systems (9), Microwave and Wireless Technology Letters (6), Electron Device Letters (4), J. Microwaves (2). **APS:** Physical Review Letters (5), Physical Review Applied (2), Physical Review Materials (2), Physical Review A (1), Physical Review B (1). **Other:** Physics of Fluids (1), Micromachines (10), Photonics (2), Sensors (1), Microwave (MDPI) (2), Progress in Quantum Electronics (1).

Selected Graduate Coursework

Wave Propagation in Elastic Solids; Advanced Vibrations of Structures; Advanced Engineering Acoustics; Theory of Solid Continua; Math Methods in ME; Advanced Solid State Microwave Circuits; Electromagnetics Theory; Introduction to MEMS; Monolithic Amplifier Circuits; Analog Integrated Circuits; VLSI Design; Computer Architecture; Digital Communication Signals and Systems; Probabilistic Methods in Engineering